1 **(a)** Define *gravitational potential at a point*.

.....

..... [1]

(b) A satellite of mass m orbits a planet of mass M and radius R_p. The radius of the orbit is R. The satellite and the planet may be considered to be point masses with their masses concentrated at their centres. They may be assumed to be isolated in space.

For the satellite in its orbit, show that

(i) its kinetic energy E_k is given by $E_k = \frac{GMm}{2R}$,

[2]

(ii) the ratio $\frac{\text{kinetic energy of satellite } E_k}{\text{total energy of satellite } E_T}$ is equal to -1.

[2]

(c) The satellite in **(b)** gradually loses energy due to small resistive forces.

Suggest why many such satellites eventually 'burn up' in the Earth's atmosphere.

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..... [3]

[Total: 8]